

Comparisons between Brains of a Large and a Small Hystricomorph Rodent: Capybara, *Hydrochoerus* and Guinea Pig, *Cavia*; Neocortical Projection Regions and Measurements of Brain Subdivisions

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Abstract. Somatic sensory, auditory and visual areas of cerebral neocortex were mapped in anesthetized capybaras using surface macroelectrode-evoked potential recording methods. The cortical motor area was mapped using electrical stimulation methods. The results of these experiments in the largest living rodent were similar to those found for the cortical sensory and motor areas of guinea pigs, a small rodent in a closely related family. The representation of the perioral skin in SI cortex was relatively large in capybaras and guinea pigs. In capybara, several cortical sulci reliably demarcate different cortical projection areas from one another.

Quantitative neuroanatomical comparisons of volumes and neuron numbers in several major prosencephalic nuclei revealed that all nuclear masses are larger in capybara than in guinea pig, but that different nuclei are enlarged to different degrees. Possible causes of larger brains in larger animals are discussed.

Introduction

Among closely related genera, larger mammals typically have larger, more convoluted and fissured brains – see reviews by JERISON [1973] and SACHER [1970]. It was the purpose of this study to gather neurophysiological and neuroanatomical data associated with brain size differences in two closely related mammals of different body size. We compared brains of capybaras, the largest living rodents, with those of guinea pigs, small rodents of a related family [CABRERA, 1957–1961; SIMPSON, 1948]. Somatic sensory, visual, auditory and motor areas of cerebral neocortex were mapped using conventional surface electrical stimulation and ma-

croelectrode-evoked potential recording methods. Individual differences in brain size and cortical fissurization were noted and quantitative neuroanatomical comparisons were made. These studies were carried out in 1959–1960 and first reported in 1960 [CAMPOS and WELKER]. Our data illustrate several features of organization of cortical sensory and motor areas in capybaras and indicate some of the problems encountered in comparative quantitative neuroanatomical studies. However, they do not provide answers to questions regarding the neurobiological significance of increased brain size.

Materials and Methods

Subjects

Ten capybaras (*Hydrochoerus hydrochoerus*) and four guinea pigs (*Cavia porcellus*) were used in the electrophysiological studies (fig. 1). Three capybaras and two guinea pigs were used for the quantitative neuroanatomical studies (fig. 11). The body weights of the capybaras ranged from 4.5–15.4 kg, and their brain weights ranged from 42.5–64.0 g. Animals of both sexes were used. Although capybaras and guinea pigs are not as closely related as are several other rodents that differ in body and brain size, we chose these two types because the capybara has the largest brain [BEDDARD, 1899; FRIANT, 1954; PILLERI, 1959], and the guinea pig is readily available. Capybaras are the largest living rodents, weighing over 50 kg when adults. They live near streams and rivers in family groups of 6 or 8. They have webbed feet and a great capacity for diving and swimming underwater for long distances.

Preparation

Conventional anesthetic (intraperitoneal sodium pentobarbital, initial dose = 25 mg/kg body weight), physiological (rectal temperature maintained between 33–37.5 °C; physiological saline injected subcutaneously), surgical and head fixation procedures were used [BENJAMIN and WELKER, 1957; LENDE and WOOLSEY, 1956; WELKER and SEIDENSTEIN, 1959; WOOLSEY *et al.*, 1952]. However, for the first time in this laboratory, an acrylic dam bonded to skull around the exposed cortex and filled with mineral oil was employed to prevent the cortical surface from drying, a known factor in promoting spreading depression. Artificial respiration via trachea cannula was routinely used for capybaras as was cortisone acetate (2.8 mg/kg body weight; intramuscular), the latter to prevent cardiac failure and vascular collapse.

Electrophysiological Mapping Methods

Stainless steel macroelectrode recording and mapping methods were similar to those used over the years by WOOLSEY and his colleagues [BENJAMIN and WELKER, 1957; LENDE and WOOLSEY, 1956; PINTO-HAMUY *et al.*, 1956; WOOLSEY, 1952]. Electrodes were 1/3 mm in diameter and surface recording loci were 1 mm or less apart in a rectangular grid pattern. The location of each electrode placement was marked on a brain photograph of the brain being studied in relation to the vascular pattern and cortical sulci. Since in the raccoon studies [WELKER and SEIDENSTEIN,

1959] we found evoked 'hash' noise responses, amplified and heard via a loudspeaker, easier to identify and more quickly discriminating of minimal field size, these evoked signals were the criterion responses used in the present somatic sensory experiments. However, evoked slow wave potentials were used in the auditory and visual experiments. The evoked potentials were recorded from the oscilloscope tube by a Grass camera in order to provide the data used in the amplitude measurements.

Sensory Stimulation

Somatic sensory. Because 'hash' sounds could be evoked by much more gentle mechanical stimulation of skin surfaces than was required to activate visible evoked slow wave potentials in somatic sensory cortex, the receptive fields delineated were much smaller than in the slow wave studies [see KRISHNAMURTI *et al.*, 1976]. The convention of defining a focus and a fringe zone for each receptive field was abandoned since this distinction was difficult to make using these criteria. Moreover, we felt that larger fields often identified by stronger mechanical stimulation were possibly due to stimulus spread through body tissues to activate distant receptors. Consequently, receptive fields were delineated by gentle touch of a tiny probe and their location, size and shape were indicated on a figurine of the body by a single contour line (see fig. 3, 4b). Five capybaras and two guinea pigs were used to define the main somatic sensory cortex (SI).

Auditory. The acoustic stimulus consisted of clicks (40- μ sec duration, 10 db SPL, 1/sec) in a free field from a 12-inch diameter loudspeaker placed 1 ft from the animal's contralateral ear. Five capybaras and one guinea pig were used in these experiments. In three experiments with capybaras, pure tone pips (50- μ sec duration) were used as stimuli (1 and 21 kHz, both at 30 db SPL) were emitted from the loudspeaker. In these cases the animals were placed in a double-walled, sound-insulated, electrically-shielded chamber.

Visual. The light stimulus consisted of a brief light flash (interstimulus interval: 1 sec) from a strobotron tube (Sylvania, SA-309) placed 3-4 inches from the contralateral eye. Two capybaras and one guinea pig were used in these experiments.

Electrical stimulation. Mapping methods using electrical stimulation of the cortical surface were those developed by WOOLSEY and his colleagues [WELKER *et al.*, 1957; WOOLSEY, 1952; WOOLSEY *et al.*, 1952]. The electrical stimulus consisted of constant current 60 Hz sine wave stimuli of 3-sec duration (interstimulus interval of 2-3 min) delivered through a monopolar stainless steel electrode (1 mm diameter). Effective stimuli ranged from 0.5 to 2.0 mA. An unresponsive (NIL) cortical point was defined as one that did not evoke a muscle movement at a current strength of 2.1 mA.

Histological Procedures

All experiments were terminated by an overdose of anesthetic followed by intracardial perfusion with 0.9% saline and formol saline. Brains were removed from the skull and photographed from standard views. Several brains were embedded in celloidin, processed histologically, sectioned, and alternate sections stained with thionine for cell bodies and hematoxylin for myelin sheaths. To assess degree of laterality of afferent projections from the eye, one eye was enucleated in one capybara and in one guinea pig and the animals were allowed to survive for 12 days. The pres-

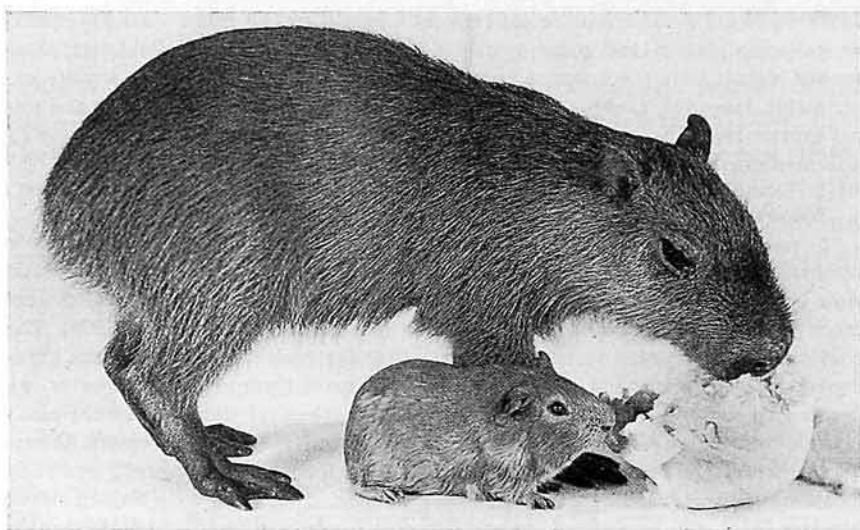


Fig. 1. Young male capybara (*Hydrochoerus hydrochoerus*, No. 62-621, body weight = 7.88 kg, brain weight = 51.0 g) and young adult male guinea pig (*Cavia porcellus*, No. 62-620, body weight = 900 g, brain weight = 4.94 g). The capybara shown weighed only about 8 kg, whereas fully mature adult males may attain a body weight of 45 kg or more.

ence of fiber degeneration in the optic tracts was checked on coronal brain sections stained with osmium tetroxide (Marchi method) which stains degenerating myelin sheaths.

Quantitative Neuroanatomy

Volumes of major nuclear groups of the prosencephalon were estimated by the methods described by DORNFELD *et al.* [1942] from planimetric measurements of drawings of enlarged projections of brain sections of one capybara and one guinea pig (fig. 11a, b) [WELKER *et al.*, 1964]. Estimates of numbers of neurons in two prosencephalic nuclear groups (fig. 11c, d) were made by the methods described by ABERCROMBIE [1946]. Brain weights were made directly of perfused specimens which included the entire brain and the first cervical segment of the spinal cord (fig. 2, 10). Body weights were also directly taken of each specimen (fig. 10). Details regarding the methods for obtaining estimates of neuron numbers in cerebral neocortex are presented pictorially and in the legend of figure 11. Cell numbers were probably somewhat overestimated due to omission of invisible cell fragments [EBBESSON and TANG, 1965] and double-counting of neurons with multiple nuclei [CARRIERE and PATTERSON, 1962]. No corrections were attempted for such possible errors nor were statistical tests made of differences in the distributions. Rather, our

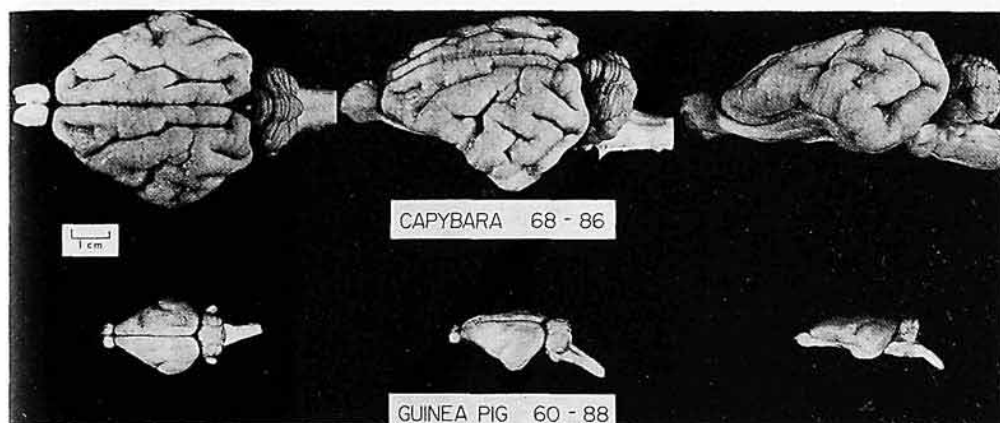


Fig. 2. Brains of a young adult capybara and an adult guinea pig from dorsal, left dorsolateral and left lateral views, pictured at same enlargement (see scale at center left). Capybara body weight: 28.2 kg; brain weight (as shown): 52.0 g. Guinea pig body weight: 0.91 kg; brain weight (as shown): 4.5 g. The largest capybara brain in our collection weighed about 64 g (see fig. 8). The largest guinea pig brain weight was about 5 g.

attempt has been merely to illustrate how measures of tissue volume or of cell number yield different results.

Results

The relative sizes of the bodies and brains of capybaras and guinea pigs are illustrated in figures 1, 2 and 10. The larger and more convoluted brain of capybara (fig. 2, 7) accords with what would be expected for a mammal of its body size [JERISON, 1973]. Our major concern was to establish the location and relative sizes and positions of the major sensory areas of cerebral neocortex, and to estimate volumes and neuron numbers of several prosencephalic nuclear groups. Our intent was that such data would constitute a preliminary step in documenting ways in which large and small brains of closely related animals differ from one another.

Neurophysiological Studies

Since the sensory and motor areas of guinea pigs have been studied and illustrated by ZEIGLER [1964], our data from this animal will not be reported here except in the composite diagram of figure 6. It is sufficient

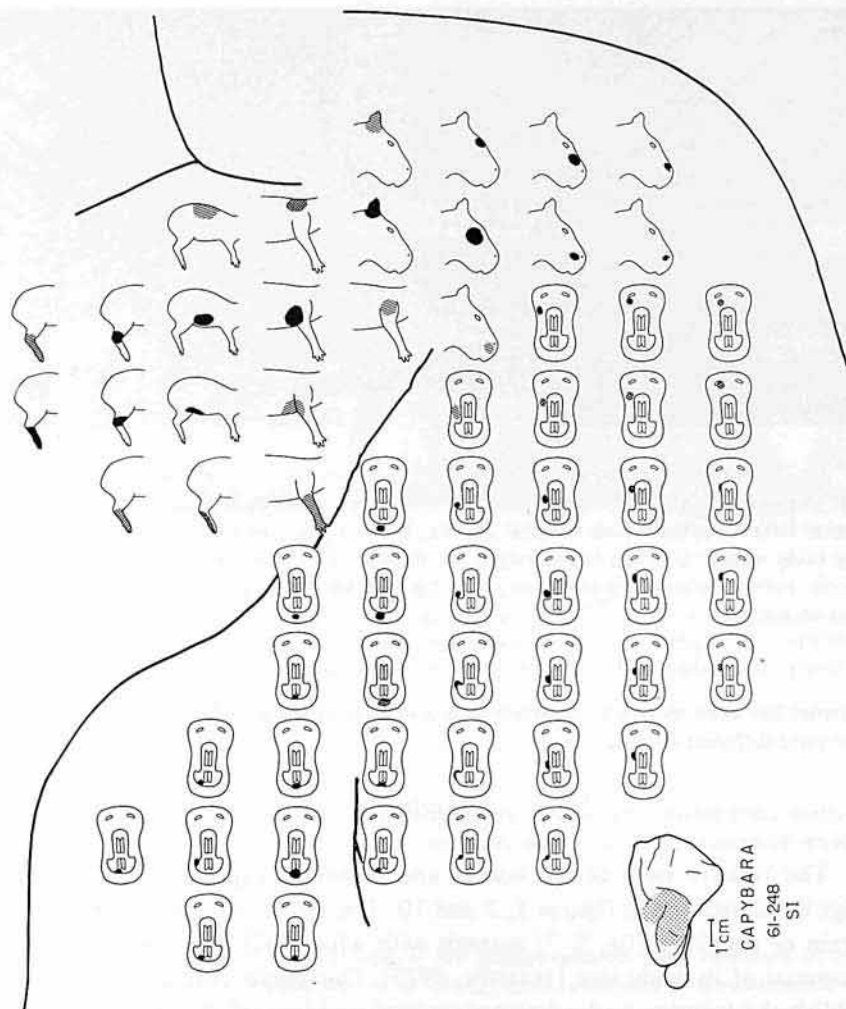


Fig. 3. Figurine map of SI, somatic sensory projections to left hemisphere of capybara 61-248. Each figurine indicates responsive cortical point. Small black patch on each figurine indicates peripheral receptive field, tactile stimulation of which evoked the maximal amplitude slow waves from that cortical point. Hatched patches signify body regions from which smaller potentials were evoked. Dotted region on small brain diagram at lower left indicates location of SI area.

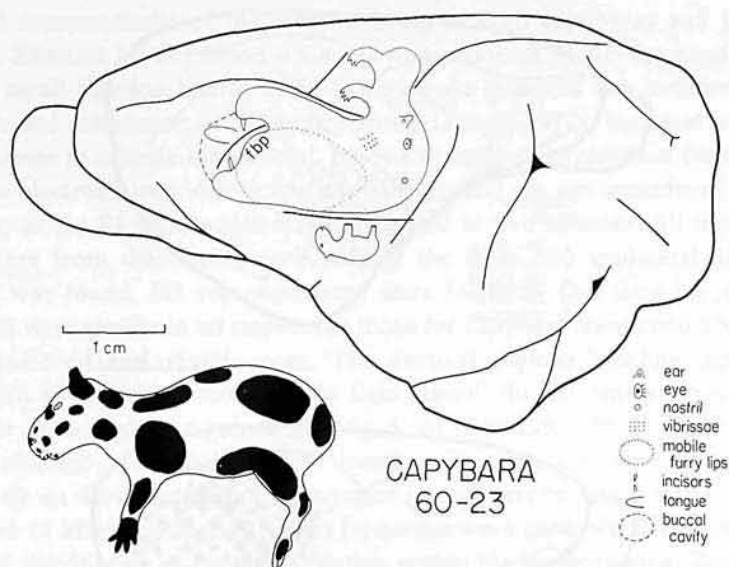


Fig. 4. *a* Schematic diagram of somatotopic pattern of projections of right body surface onto left cerebral neocortex (SI), showing relative sizes of cortical projections of different body regions (see keyed symbols at lower right), and relationships of sulci to these projections. Note relative locations of nostril, eye, ear, mystacial vibrissae, mobile furry lips and furry buccal pad (fbp) within the relatively large head region. Small figurine lateral to head area indicates location and suggested orientation of SII area. *b* Outline drawing of left lateral side of body of capybara showing relative sizes of receptive fields that evoked maximal audible 'hash' responses at single cortical surface locations within appropriate portions of somatotopic pattern of SI. Note smallest fields from lip regions and largest fields on trunk. Stimuli consisted of gentle mechanical stimulation.

to indicate that our results regarding the locations, size and orientation of the sensory areas of guinea pigs are essentially identical to those obtained by ZEIGLER. We did not study the motor area of guinea pigs.

Somatic sensory areas. The pattern of organization of SI cortex is illustrated in figure 3 by the figurine map from one experiment. Figure 4a shows the locations of SI and SII of capybara. The SI figurine illustrates the orientation and pattern of organization of projections from mechanosensory receptive fields on the right side of the animal's body to cortex of its contralateral (left) cerebral hemisphere. It is clear that SI is somatotopically organized but is distorted by differential enlargement of projec-

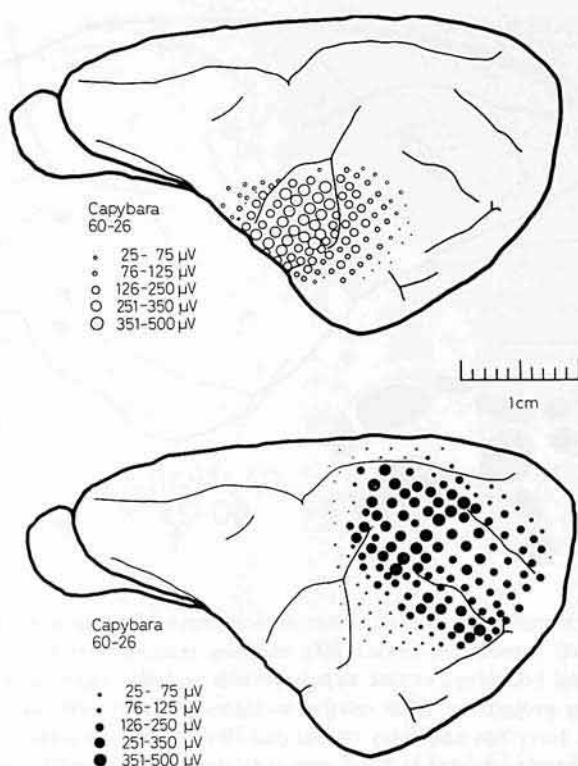


Fig. 5. Diagrams of dorsolateral view of left hemisphere (capybara No. 60-26) showing locations of auditory (top) and visual (bottom) regions. Each circle or dot on the brain diagrams indicates a recording locus. Larger circles denote greater amplitudes of surface positive-evoked potentials as indicated in the amplitude key to the left of each drawing. Smallest circles represent evoked slow waves of smallest amplitude, and dots indicate NIL points. Auditory stimuli were click stimuli delivered to the contralateral ear from a loudspeaker. Visual stimuli were light flashes to the contralateral eye.

tions from certain body parts, particularly those that are utilized in this animal's [CABRERA and YEPES, 1940; CRANDALL, 1964; PALMER, 1957; WALKER *et al.*, 1964] discriminative browsing and grazing habits. The largest representation, relatively, is that of the furry buccal pad and the surrounding grasping mobile lip areas of the muzzle. Moreover, the sizes of the receptive fields from these mouth parts are the smallest (fig. 4b). The head representation constitutes about 80% of SI by surface

areal measurements of the mapped brain in both capybaras and guinea pigs. ZEIGLER [1964] found a similar proportion of SI for the head area. The small figurine lateral to SI in figure 4a indicates the location, and presumed orientation of SII projections [HAIGHT, 1972], but clear evoked responses to stimulation of head, limbs and body were obtained from only a few electrode locations in one experiment, and are not considered as reliable as the SI data, which were replicated in five animals. All inputs to SI were from the contralateral side of the body. No ipsilateral SI face area was found. SII representations were bilateral. Our data for guinea pig SI were similar in all respects to those for capybara mentioned above.

Auditory and visual areas. The cortical regions yielding maximal evoked responses to acoustic and light stimuli do not appear to overlap either in capybara or guinea pig (fig. 5, 6) [ZEIGLER, 1964]. These areas are situated in similar cortical locations in capybara and guinea pig (fig. 6). In the one auditory experiment in capybara in which tonal stimuli of low (1 kHz) and high (21 kHz) frequency were used, we found a differential distribution of evoked potentials within the auditory area. The cortical locations of greatest response amplitude to the low frequency tone were 5–6 mm caudal to those responding to the high frequency tone. Spatial separation of different frequencies within auditory cortex has also been demonstrated in rabbits as well as in other mammals [WOOLSEY, 1971]. Visual-evoked potentials were contralateral to the eye stimulated. Ipsilateral stimulation was tried at a few cortical locations where contralateral-evoked responses were pronounced, but no potentials were evoked in these instances. This result is supported by our finding of degeneration (Marchi method) only in the contralateral optic tract following unilateral enucleation of one eye in both capybara and guinea pig.

It should be mentioned that the boundaries of sensory areas cannot be accurately determined by macroelectrode-evoked slow wave methods due to volume conduction artifacts [WELKER and SEIDENSTEIN, 1959]. Several studies have shown that the volume-conduction artifact can be minimized [PERL and CASBY, 1954] by use of microelectrodes instead of macroelectrodes [KRISHNAMURTI *et al.*, 1976; MOUNTCASTLE and HENNEMAN, 1952]. For this reason we have taken the conservative view that the three major sensory areas do not overlap with one another (fig. 6), although small evoked potentials do overlap between the areas in certain locations (fig. 4). Since small cortical 'association' areas are situated between the major sensory regions in rodents as well as in other mammals it is expected that neurons in such areas will be found that respond to multimodal in-

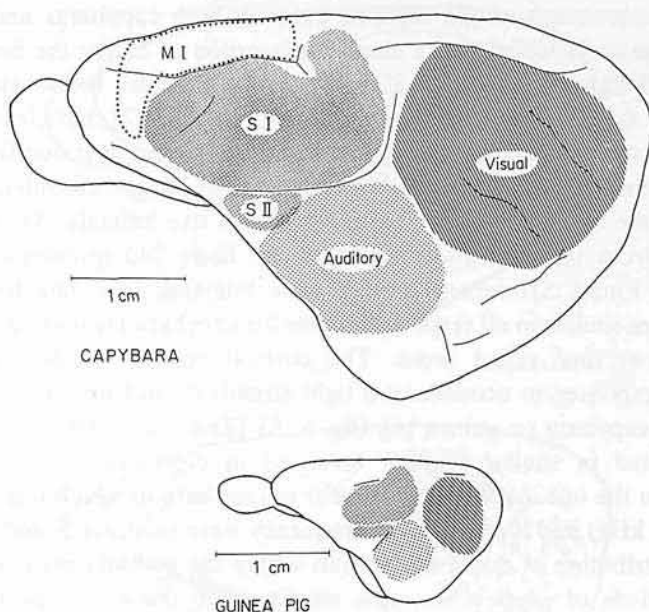
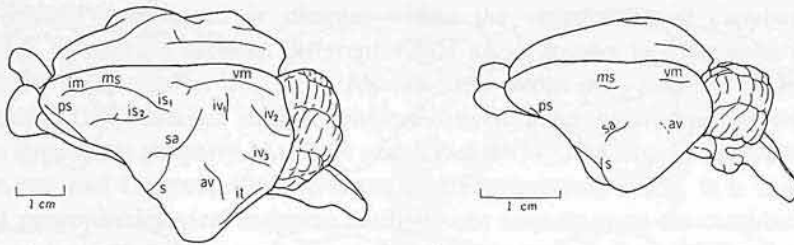


Fig. 6. Comparison of sensory areas of capybara (top) and guinea pig (bottom) showing locations of somatic sensory (SI and SII), auditory and visual areas on composite brain diagrams, drawn to same scale. Shaded symbols for sensory areas are same for capybara and guinea pig. Motor area (MI), defined in one capybara experiment, is indicated in region surrounded by large dots. Electrical stimulation within this region elicited muscle movements. Muscles of trunk, limbs and neck were activated by stimulation of cortex medial to the longitudinally oriented sulcus (ms, see fig. 7), and those of the face, and especially the upper lip and furry buccal pad, were activated from cortex lateral to this sulcus rostral to SI. Lip movements were also elicited from stimulation of the rostral portion of SI comprising the sensory representation of the lip and perioral skin (see fig. 3, 4), but are not included in this schematic delineation of MI. Muscles activated were contralateral to the side of the cortex stimulated.

puts. Microelectrode mapping methods can help to answer this question definitively. [PIMENTEL DE SOUSA *et al.*, in preparation].

Motor area. Cortical stimulation was carried out in one capybara. The regions explored included the entire SI area and cortex rostral to SI as well as medial to it across sulcus ms (fig. 7b) as far as the midline (fig. 6). Muscle movements were elicited reliably from cortex medial to sulci ms and im and rostral to SI, as well as from rostral portions of SI comprising



Capybara 61-248

Guinea Pig 62-620

Fig. 7. Diagrams of capybara (left) and guinea pig (right) brains (not drawn to same scale) showing sulci, most of which have been named according to their relative position with respect to physiologically, and/or anatomically, defined boundaries. Abbreviations: ps, presylvian or orbital sulcus (named for its similar location and orientation to that of primates and carnivores); ms (motor-sensory sulcus, homologous with central sulcus in primates or postcruciate sulcus in carnivores). The rostral portion of this longitudinal sulcus (im, intramotor sulcus) lies between the body portion of 'motor' (MI) cortex medially and the face and lip portion of MI laterally; is₁ (intrasomatic sensory sulcus, homologous with coronal sulcus in carnivores), lies between body and face subdivisions of SI; is₂ (intrasomatic sensory sulcus), lies partly between upper and lower lip representations and partly between upper lip and buccal representations; sa (somatic-auditory sulcus, homologous with anterior suprasylvian sulcus in carnivores), lies between the head area of SI and the auditory region. The rostral spur of sa lies between SI and SII in this brain; s, sylvian sulcus, standard location and orientation for mammals; vm (medial visual sulcus, usually identified as the splenial sulcus in rodents), lies at the medial boundary of the visual region; sulci iv₁₋₃ (intravisceral sulci), lie within the region responsive to visual stimuli; av (auditory-visual sulcus), lies between the auditory and visual regions; it, infero-temporal sulcus, arbitrarily named for its relative position and orientation similar to that in other mammals.

the entire perioral sensory representations. Face and perioral motor representations were found only in the perioral region of SI and in cortex rostral to this portion of SI. Motor representations of musculature of the neck and the rest of the body were situated only medial to sulci ms and im up to the midline. Midline cortex was not explored. The boundaries of the motor area are difficult to determine with the macroelectrode surface stimulation method due to the possibility of stimulus current spread to distant cortical regions as well as to underlying fibers of passage from distant regions [WELKER *et al.*, 1957]. The fact that thresholds were generally rather high (1.5 mA and above) throughout the area stimulated would tend to further result in spread of the stimulus. Our limited data do not

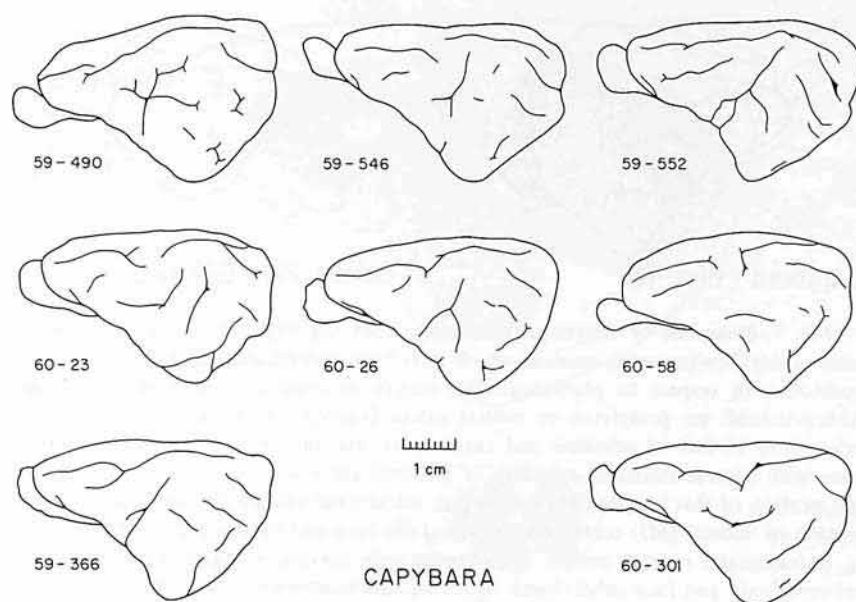


Fig. 8. Drawings of left dorsolateral views of brains of eight capybaras illustrating variations in sulcal patterns in different specimens.

show conclusively that the facial motor responses elicited from rostral SI cortex constitute the motor outflow of SI, since electrical thresholds for production of movements were not higher there than they were in MI cortex situated more rostrally. Examination of cortical cytoarchitecture in this region reveals that sensory cortex with a well developed 4th layer is coincident with the SI map and cortex with large pyramidal cells in the 5th layer lies both rostral to SI and medial to sulci ms and im [fig. 6, 7, 9). At any rate, the difference in locations of the motor representations of face and body is valid. We did not attempt to define a supplementary motor area [WOOLSEY, 1958].

Fissurization and functional correlations. Capybara cortex has several fissures and sulci whereas cortex of guinea pig exhibits only a few tiny dimple-like sulcal depressions (fig. 2, 7). Our neurophysiological data revealed that several sulci in capybara, and the dimples in guinea pig, appear to lie between either different sensory areas, a sensory and a nonsensory area, SI and MI, or different projectional subdivisions within SI or MI (in capybara). These areal-sulcal correlates are shown in figures 3 and

5. Several small sulci or dimples within the visual area of capybara (fig. 5, 6) may lie between different visual areas, known to exist even in rodents [MONTERO *et al.*, 1973, for rats, and WOOLSEY *et al.*, 1973, for rabbits]. Only detailed in-depth *microelectrode* mapping studies can test this hypothesis properly [ALLMAN and KAAS, 1974; JOHNSON *et al.*, 1973; WELKER and CAMPOS, 1963; WELKER and SEIDENSTEIN, 1959]. It is clear that neurophysiological mapping methods can contribute to the establishment of a consistent nomenclature for homologous cortical fissures and sulci in different animal types. Our suggestions for a consistent kind of functional nomenclature for sulci in capybaras and guinea pigs are shown in figure 7. Interanimal variations in details of sulcal pattern do occur, however (fig. 8), and appear considerable in capybara [ELLIOT SMITH, 1902; BEDDARD, 1899; PILLERI, 1959]. Even so, the larger, deeper cortical fissures appear to be consistently situated in capybaras, as are the dimples in guinea pigs. In some capybaras the long longitudinally oriented composite sulcus near the midline (including im, ms, is and vm) is disrupted such that a gyral bridge appears (e.g. figs. 4a, 6, 7a, 8). In every case where this region was studied electrophysiologically, this fissural gap contains the body representations of SI. In those cases where the gyral bridge is absent, the body representations lie buried within both walls of the sulcus. We believe that interspecies as well as species differences in sulcal pattern reveal real differences in orientation, relative size and location of different but adjacent functional projections [WELKER and SEIDENSTEIN, 1959; WELKER and CAMPOS, 1963].

Neuroanatomical Studies

Cytoarchitecture. The cytoarchitecture of somatic sensory, auditory and visual cortex is distinct from surrounding cortex in both capybara and guinea pig (fig. 9). The cytoarchitecture of motor cortex also is distinct (fig. 9) with respect to classically defined characteristics [BRODMANN, 1909; ROSE, 1942; SANIDES, 1962]. However, we have not attempted to demonstrate a precise correspondence of the boundaries of these areas as defined by the electrophysiological and cytoarchitectural criteria due to the limits imposed by macroelectrode surface recording methods as well as by the variability that characterizes descriptive methods of delimiting cytoarchitectonic areas and their boundaries [LASHLEY and CLARK, 1946].

Quantitative neuroanatomy. Our quantitative data, summarized in table I and figure 11, although derived from only one capybara and one

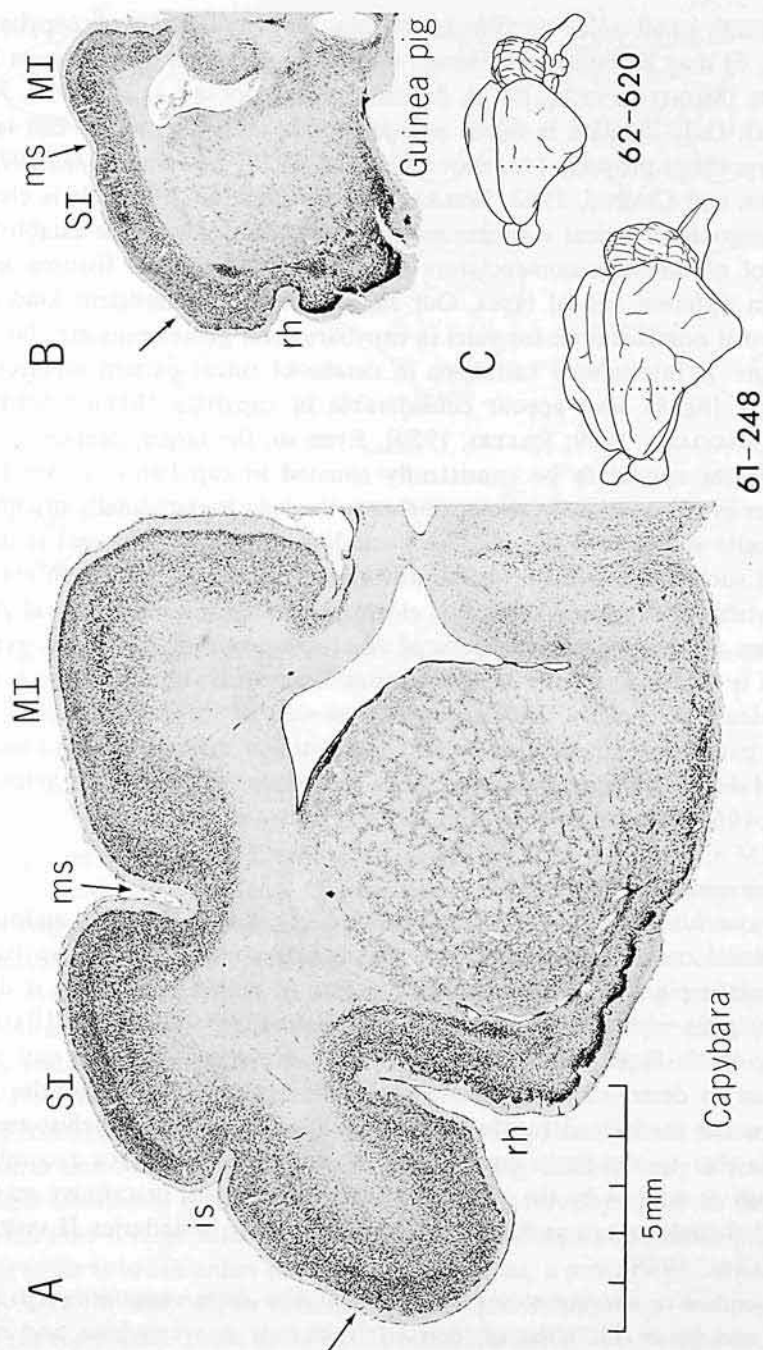


Fig. 9

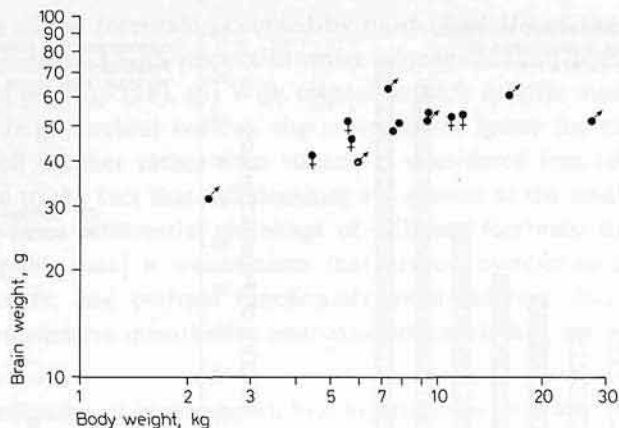


Fig. 10. Body weights and brain weights of 13 capybaras plotted on logarithmic scales. Sex of each animal indicated by standard symbols or by a dot if the sex was not recorded.

guinea pig, reveal several simple results, all of which might be expected (see Discussion). (1) All capybara forebrain structures are larger than comparable guinea pig structures. (2) With respect to both volume and cell number, different nuclear groups of the prosencephalon are not equally enlarged or greater in capybara when compared with those of guinea pig. (3) There is an orderly sequence of this differential enlargement of the several major forebrain structures: the cerebral cortex is enlarged most, structures related to it are less enlarged and basal telencephalic and other comparatively more conservative structures are least enlarged in capybara when compared with guinea pig (fig. 11a). (4) The

Fig. 9. Coronal sections of left cerebral hemisphere of capybara (A) and guinea pig (B). Thionin stain. *A* Section (No. 612) through sulcus *ms* (medially) and *is* (laterally) of capybara No. 61-248. Cytoarchitecture medial to *ms* is typical of motor cortex (MI) with large pyramidal cells in Vth layer. Cytoarchitecture lateral to *ms* and medial and lateral to *is* is typical of somatic sensory cortex (SI) with well developed IVth layer. *rh* = Rhinal sulcus. *B* Section (No. 630) through motor and sensory cortex of guinea pig No. 62-620 at about the same rostrocaudal level as in capybara section shown in A. Arrow at *ms* indicates boundary between motor (MI) and somatic sensory (SI) areas, and coincides with location of slight cortical 'dimple' regularly found at this cytoarchitectural boundary. *C* Brain diagrams of capybara (left) and guinea pig (right), with dotted lines indicating locations of the sections in A and B. In both A and B, SI cortex lies between the two arrows. Both sections magnified to same scale.

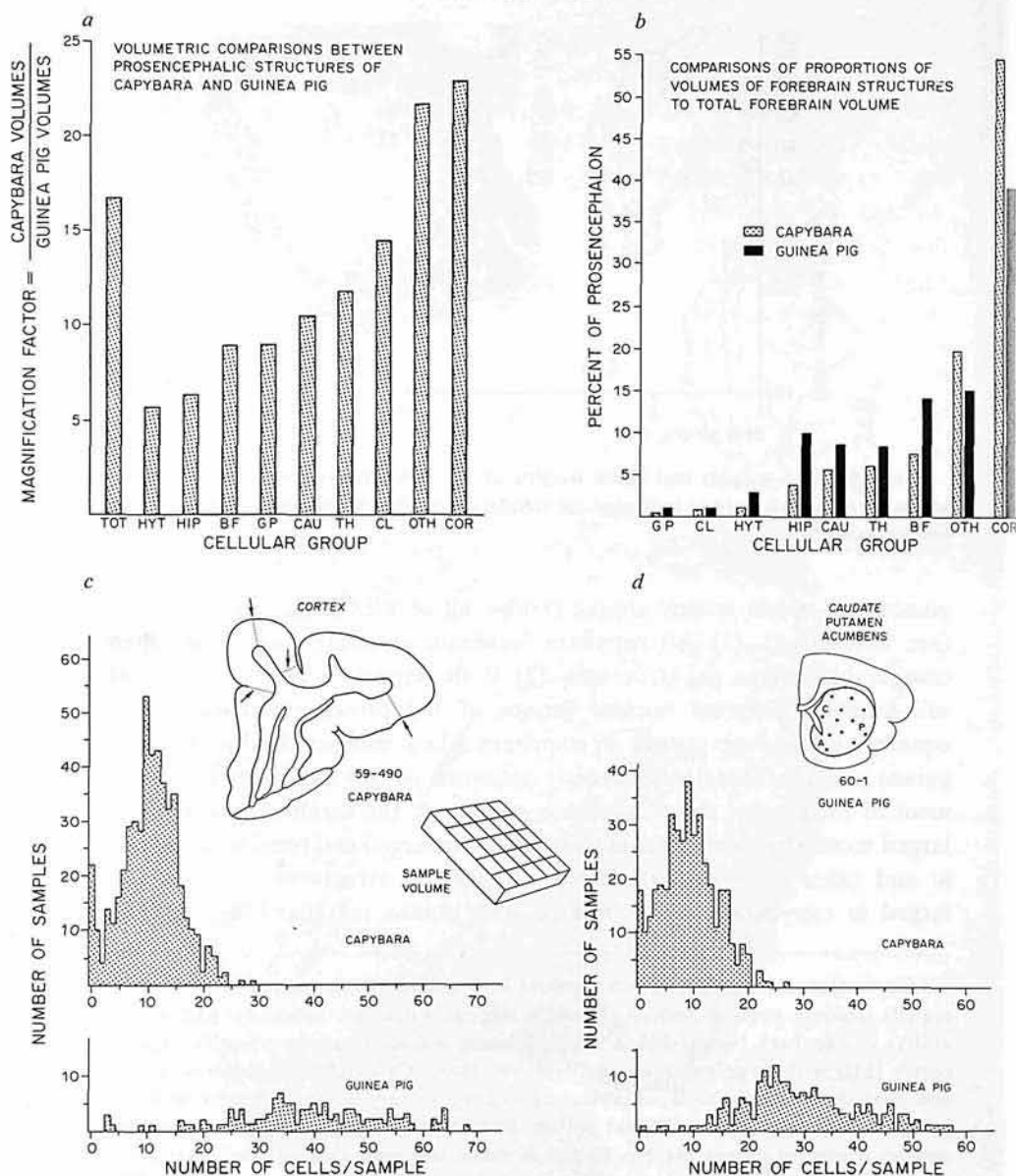


Fig. 11. Quantitative neuroanatomical comparisons between capybara (No. 54-490) and guinea pig (No. 60-1). *a* Number of times capybara forebrain structures are larger than those of guinea pig. Abbreviations: cau, caudate, putamen and accumbens complex; cl, dorsal claustrum; cor, cerebral neocortex; gp, globus pallidus; hip, hippocampus (including ammons horn, subiculum, dentate gyrus and

proportion of the forebrain occupied by most (8 of 10) of the structures measured is similar with respect to order of magnitude in both capybara and guinea pig (fig. 11b). (5) With respect to each specific nuclear group measured (e.g. cerebral cortex), the enlargement factor for capybara is lesser if cell number rather than volume is considered (see table I). (6) This is due to the fact that cell densities are greater in the smaller guinea pig brain. Since differential shrinkage of different forebrain structures is likely [unpubl. data] it would seem that neuron number would be the more accurate, and perhaps functionally more relevant, statistic to be used in comparative quantitative neuroanatomical studies, since cell num-

anterior continuation of hippocampus); hyp, hypothalamus (including hypothalamic nuclei and inclusive fiber bundles); bf, basal forebrain (including anterior olfactory nuclei, tuberculum olfactorium, pyriform and prepyriform cortex, entorhinal cortex, diagonal band of Broca, septal nuclei, subfornical organ, bed nuclei of stria terminalis and of anterior commissure, amygdaloid nuclei, periamygdaloid cortex); oth, other structures (refers to all major prosencephalic fiber tracts: internal capsule, external capsule, corpus callosum, subadjacent cortical fibers, capsula extrema, anterior commissure, fibers within pyriform lobe and peduncle); th, thalamus (dorsal thalamus, epithalamus and ventral thalamus, including fiber tracts but excluding pretectal nuclei); tot, total prosencephalon (includes all of the above as well as miscellaneous small fiber tracts not otherwise measured). *b* Comparison of capybara and guinea pig with respect to percent of prosencephalic volume occupied by each of several major forebrain structures. Abbreviations same as in *a*. *c* Distribution of cell densities of cerebral neocortex in capybara (above) and guinea pig (below). The size ($90 \times 90 \times 20 \mu\text{m} = 162,000 \mu\text{m}^3$) and shape of the sample volume used is shown in the middle of the figure. All neurons containing nucleoli were counted in a series of nonoverlapping, but adjacent, samples from cortical surface to depth of the cortex at representative locations throughout the right hemisphere, oriented as indicated by the dotted lines in cortex (indicated by arrows) in drawing at left, of right coronal section of forebrain (capybara, section No. 771). Total number of cortical locations sampled = 29 for capybara and 14 for guinea pig. Total number of cortical samples = capybara, 496; guinea pig, 124. Total number of cortical neurons counted = capybara, 5107; guinea pig, 4778. *d* Distribution of cell densities of caudate, putamen and accumbens nuclear complex in capybara (above) and guinea pig (below). The size and shape of the sample volume used was the same as for cortex. However, at each location, 4 adjacent samples were taken. All neurons in these four adjacent samples were counted in irregularly spaced sample locations throughout this nuclear complex as indicated by large dots in drawing at right, of a right cross-section of guinea pig forebrain (section No. 531). Total number of locations sampled = 50 for capybara, 22 for guinea pig. Total number of samples = capybara, 408; guinea pig, 201. Total number of neurons counted = capybara, 3,741; guinea pig, 5,894. Volumetric measurements in *a* and *b* were made in the right hemisphere of both Capybara and guinea pig.

Table I. Quantitative comparisons of brains of capybara and guinea pig

	Volumes, mm ³									Cortico-thalamic ratio
	total pros.	neo-cortex	white	thal	cau	gp	bf	hip	hy	
Capybara No. 59-490	7,365.0	3,979.0	1,451.0	434.1	413.0	41.0	551.2	278.6	74.6	× 9.17
Guinea pig No. 60-1	443.1	170.2	96.26	36.6	37.7	4.5	61.5	45.5	13.1	× 5.15
Multiplication factor ³	16.62	23.38	15.07	11.86	10.95	9.11	8.96	6.12	5.69	1.78

	Areas, mm ²		Percent cortex in fissures	Cortical thickness
	total cortex surface	cortex in fissures		
Capybara No. 59-490	2,501.0	760.3	30.4	1.59
Guinea pig No. 60-1	218.2	7.0	3.2	0.78
Multiplication factor ³	11.46	108.61	9.5	2.04

	Mean number of cells/sample ¹		Cell densities ² number of cells/mm ³		Total cell numbers	
	cortex	cau	cortex	cau	cortex	cau
Capybara No. 59-490	10.3	9.169	63,582	56,600	252,992,778	23,375,800
Guinea pig No. 60-1	38.53	29.32	237,846	180,992	40,481,389	6,823,398
Multiplication factor ³	3.74 ³	3.0 ³	3.74 ³	3.0 ³	6.25	3.43

Abbreviations: same as for figure 11. Pros = Prosencephalon.

¹ Sample volume for all density counts = 162,000 μm^3 ($90 \times 90 \times 20 \mu\text{m}$).

² Number of sample volumes per mm^3 = 6,173.

³ The multiplication factors for all data (except number of cells per sample and cell densities) refer to the number of times the capybara structure (or cell number) is greater than that of guinea pig.

ber is not affected by differential shrinkage. (7) The proportion of surface area of cerebral cortex occupied by the three sensory areas is similar in the two animals. Thus, there is no evidence that the larger capybara has relatively greater amounts of nonsensory (or association) cortex. (8) Cerebral neocortex of capybara is more fissured than that of guinea pig. Combined with the result of '7' above, the greater fissuration of capybara cortex does not appear to confer upon it proportionally more nonsensory cortex.

In general, we conclude that the questions we have asked regarding the functional significance of larger brains in larger animals are still too general. Numerous anatomical and physiological features are different in brains of different size (see Discussion). More specific hypotheses are required regarding such differences before more definitive conclusions can be reached.

Discussion

There is an extensive literature that documents the fact that larger mammals have larger brains and that increased size during evolution appears to have favored adaptive survival [GOULD, 1966; JERISON, 1973; SACHER, 1970]. This relationship is a systematic and regular one and is somewhat different for different mammalian orders. Increased fissuration of larger brains has also been noted for larger animals [CONNOLLY, 1950]. There is an even more extensive literature that attempts to associate these brain-body relationships to structural-functional causes and to their behavioral effects [JERISON, 1973; RENSCH, 1954, 1956, 1960]. Larger brains are believed to have relatively more association cortex [SANIDES, 1962], a greater number of functional cortical areas, greater cell diversity, etc., all of which contribute to increased diversity and adaptability of the behavioral repertoire, discriminative capacity, motor specialization, less dependence on fixed instinctual behaviors and more on learning, memory, intelligence and other mental complexities [WELKER, 1976]. Quantitative neuronanatomical studies of microscopic structural features of smaller and larger brains of closely related species have attempted to provide more definitive data that might explain some aspects of the functional significance of larger brains [BLINKOV and GLEZER, 1968; BOK, 1959; BOK and VAN ERP TAALMAN KIP, 1939; HARDE, 1955; SACHER, 1970; SCHULZ, 1951; SHOLL, 1948]. For example, larger brains have a greater number and variety of neurons with larger dendritic trees that presumably accom-

moderate greater numbers and varieties of synaptic connections [BOK, 1936, 1959; SHOLL, 1956].

Although several general hypotheses of the adaptive value of larger brains and of the underlying structural-functional causes have been proposed from the inception of interest in this topic, few have been subjected to experimental test, and most are very general or unrealistically simplistic [BOK, 1959; JERISON, 1973; RENSCH, 1960; SACHER and STAF-FELDT, 1974; WELKER, 1976]. It is becoming increasingly clear that numerous alterations occur in almost all organs and tissues of larger animals [ALTMAN and DITTMER, 1972; RENSCH, 1960]. It seems most likely that each specific altered structure within larger creatures would play a partial role in improving or optimising adaptive success [DEBEER, 1958; Goss, 1964; ROSEN, 1967]. In addition, not all structures are equally enlarged or more numerous in larger brains of closely related animals [HARDE, 1955; GOULD, 1966; RENSCH, 1954, 1960; SCHULZ, 1951; SHOLL, 1948]. Also, numerous microscopic structural and biochemical [MALLOUK, 1975] features of neural circuits are likely differentially altered in larger animals [SPINA FRANCA NETTO, 1951] rendering any single-factor hypothesis pitifully inadequate. SACHER [1970, 1976] proposes that an entirely different look at available data may clarify the problem in view of the fact that previous efforts have not been fruitful in establishing reliable relationships between indices of cephalization and neurobehavioral performance. He has reanalyzed data that indicate that increased brain and body sizes are also associated with longevity and measures of metabolic activity [SACHER, 1959, 1975, 1976]. It is obvious that multiple structural and functional features must be studied in a carefully designed extensive program of experimental studies before a fundamental set of data is available to test even the simplest of multiple-factor conceptions of improved adaptive functions of larger bodies and brains [HOLLOWAY, 1966a, b; WELKER, 1976].

These larger issues aside, the present study has demonstrated differential enlargement of somatic sensory cortical projections from specific body surfaces (perioral and intraoral buccal surfaces) that are actively utilized in exploratory contact with objects and surfaces (similar differentially enlarged perioral and buccal surfaces are seen in SI of sheep [JOHNSON *et al.*, 1974]. Our data reveal another variation of this general principle in that the lip and perioral projection of capybara and guinea pig are differentially enlarged whereas in other rodents, e.g. rats [WELKER, 1971], the mystacial vibrissae projections are those that are enlarged. We have

also provided additional evidence that sulci lie at boundaries between cortical convolutions with projections from sources that differ in some set of functional-structural features [WELKER and CAMPOS, 1963]. Electrophysiological mapping methods have established these principles, but progressive refinement of these methods is necessary to improve the resolution of the maps. Quantitative neuroanatomical methods have likewise established certain generalizations regarding organization of brains of different mammals [BLINKOV and GLEZER, 1968; BOK, 1936, 1959; BOK and VAN ERP TAALMAN KIP, 1939; SHOLL, 1948]. But these also require considerable refinement and diversification before they can substantially contribute to understanding of the functional significance of specific quantitative aspects of the microfeatures of neural circuits.

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